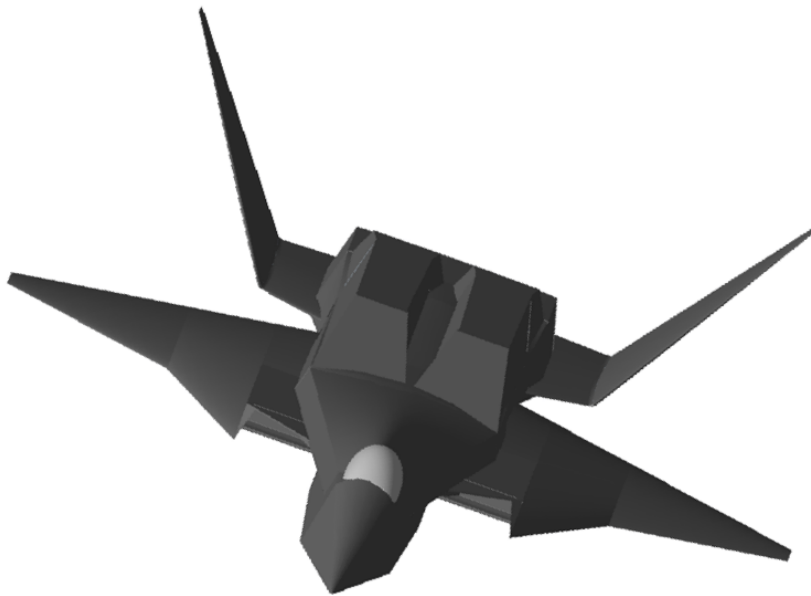

Next Generation Fighter

Individual Technical Report - GDP 2019



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Abstract

Here, the preliminary design and sizing of the air intake and exhaust system for a supersonic capable, stealth fighter aircraft is performed. An optimal inlet shock system was found and a stealth diffuser duct was designed. The air intake system gave good pressure recovery across the flight envelope, $\eta_r > 0.9$, exceeding military requirements. Optimal flight conditions are $M_\infty = 0.75$ and $M_\infty = 2.25$. The exhaust nozzle has variable geometry and is capable of pitch thrust vectoring. Stealth technologies were employed to reduce the heat signature of the exhaust gases. Methods of estimating installation losses are discussed and performed using the software GasTurb [1]. Both intake and exhaust systems are integrated into the airframe of a Next Generation Fighter concept.

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List of Symbols

A	Area	$[m^2]$
C_D	Nozzle discharge coefficient	
C_V	Nozzle velocity coefficient	
C_{fg}	Nozzle gross thrust coefficient	
F	Uninstalled thrust	$[N]$
L	Length	$[m]$
M	Mach number	$[-]$
MFP	Mass flow parameter	
N	Total number of shocks	
P	Pressure	$[Nm^{-2}]$
T	Installed thrust	$[N]$
T	Temperature	$[K]$
$\dot{m}$	Mass flow rate	$[kgs^{-1}]$
ρ	Density	$[kgm^{-3}]$
h	Height	$[m]$

Abbreviations

BLB	Boundary Layer Bleed
CAD	Computer Aided Design
CFD	Computational Fluid Dynamics
CMP	Critical Mission Point
DMP	Design Mission Profile
ECA	Engine Cycle Analysis
LERX	Leading Edge Re-extension Device
MFR	Mass Flow Ratio
NGF	Next Generation Fighter
RCS	Radar Cross Section

Greek

α	Angle of attack	$[^\circ]$
β	Ramp half angle relative to aircraft datum, nozzle divergence angle	$[^\circ]$

δ	Flow turning angle relative to upstream flow	$[\circ]$
η_r	Intake pressure recovery	
γ	Ratio of specific heats	
Ω	Duct offset ratio	
ϕ	Installation loss coefficient	
θ	Shock angle, Nozzle convergence angle	$[\circ]$

Subscripts

0 – 6	Inlet station points
7 – 9	Nozzle station points
∞	Freestream conditions
a	Actual condition or property
ava	Available
des	Design condition or property
i	Ideal condition or property
req	Required
sub	Subsonic
sup	Supersonic
T	Total
t	Throat
x	Upstream of shock
y	Downstream of shock

Superscripts

*	Choked conditions
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1 Introduction

The preliminary design process for an intake and exhaust system for a next generation fighter aircraft are covered here. Initial conceptual design, on-design and off-design sizing and resizing and consideration of installation losses are included. Station numbers referenced are given in figure 1.

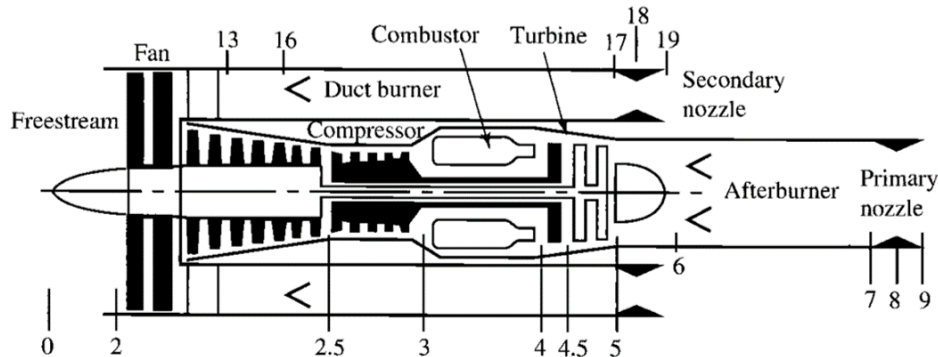


Figure 1: Gas turbine engine station numbering [2]

1.1 Intake System

Air intakes convert flow at freestream conditions across the flight envelope to conditions for optimal engine performance at the fan face. Efficient intakes maximise the conservation of total pressure. For subsonic flight this is achieved by maintaining near isentropic flow and ensuring the inlet throat is large enough not to choke the flow. For supersonic flight, air must be brought to subsonic conditions before isentropic deceleration. Multiple weak shocks give better pressure recovery than a single strong shock, so a system of oblique shocks is implemented. The section of the intake which induces shocks is referred to as the *supersonic inlet* and the section responsible for isentropic expansion is the *subsonic diffuser*. The narrowest part of the intake is the *throat*. The intake must also deliver air with low distortion [3], this is complicated by the requirement for stealth as a straight, unhindered subsonic diffuser is not an option.

1.2 Exhaust Nozzle

Maximum thrust is achieved when the exhaust static pressure matches the freestream. As the static pressure leaving the engine varies with flight conditions and throttle settings a variable geometry nozzle is required for optimal performance. A converging-diverging nozzle was used to accelerate exhaust flow to supersonic velocities. An efficient nozzle must do this near isentropically to minimise losses, restricting the maximum flow expansion and contraction angles. The narrowest part of the nozzle is the *throat* and is sized such that the flow is just sonic at this point.

2 Preliminary Design

2.1 Air Intake

2.1.1 Placing the Air Intake

Placement of the intake demonstrates the importance of collaboration between sub-teams. Initially, intakes were placed above the wings, primarily for improved stealth (appendix ??). This gave poor performance at high α and disturbed flow over the wing. Aerodynamics, structures, systems and propulsion teams had competing priorities for the intake placement. Upon discussion, intakes were re-placed within the LERX (figure 2b) preserving flow over the wing and stealth characteristics whilst avoiding wing spars and the nose gear.



Figure 2: Potential intake location with respect to wing

2.1.2 Stealth Characteristics

Rotating fan blades reflect radar in all directions and will increase radar cross section if visible. A serpentine duct shaped subsonic diffuser is the primary way of shielding the fan. Line of sight must be removed for any pitch or yaw angle. Intakes are swept to match the wing sweep, minimising the number of unique angles [4]. A square or rectangular intake is capable of always reflecting radar back to the source, skewing the intake face will disperse radar.

2.1.3 Boundary Layer Control

Boundary layer flow must be removed from the engine flow through boundary layer bleed systems. Splitter plates, used on the F-22 (figure 3a) can separate the intake from the fuselage such that the boundary layer is not ingested. This increases frontal radar cross section (RCS). A stealthier bleed system was used by the YF-23 (figure 3b). A perforated panel situated upstream of the inlet vents boundary layer air away from the intake without compromising stealth [5].



(a) F22 Raptor side mounted inlet [6]



(b) YF-23 Black Widow II under-wing inlet [7]

Figure 3: Boundary layer control systems employed on fifth generation fighter aircraft

2.2 Exhaust Nozzle

2.2.1 Stealth Characteristics

A rectangular nozzle (figures 4a and 4c) offers better stealth performance than a conventional, circular nozzle (figure 4b) as it is less likely to reflect radar back to the source. This configuration is heavier and more costly than a circular nozzle, as the rectangular plates resist pressure loads less effectively. Jagged edges, figure 4, reduce the RCS visible to any enemy directly behind the aircraft without significant contribution to weight or cost.

The YF-23 shown in figure 4c used the fuselage and tail surfaces to shield the exhaust IR signature from the side and beneath the aircraft. Serpentine ducts are used to reduce the infrared signature [11] from the rear of the aircraft and obscure high RCS contributing turbine blades.

2.2.2 Thrust Vectoring

A thrust vectoring nozzle was desired by the control team to give improved manoeuvrability and performance at high angles of attack. The majority of modern fighter jets use a circular thrust vec-

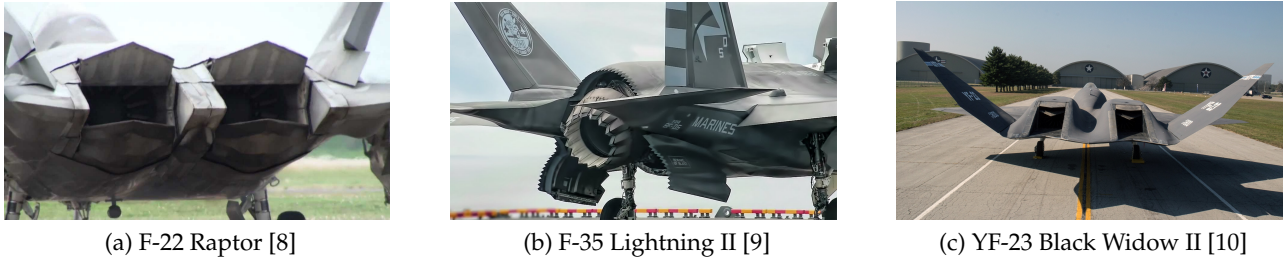


Figure 4: Exhaust nozzle design for fifth generation stealth fighter aircraft

toring nozzle. Only the F-22 has a rectangular thrust vectoring nozzle - the YF-23 sacrificed thrust vectoring in favour of improved stealth.

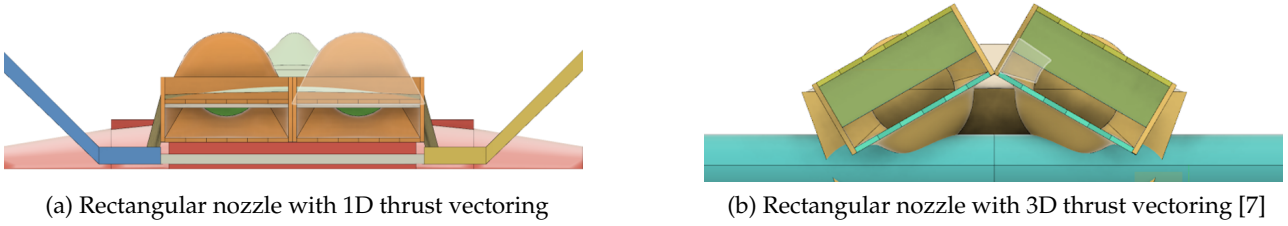


Figure 5: Thrust vectoring with rectangular nozzles

Figure 5a shows a nozzle configuration that gives pitch-only thrust vectoring, similar to that used by the F22. Figure 5b shows how two rectangular nozzles could be configured to give pitch and coupled roll and yaw vectoring. These options were proposed to the sub-team leaders and it was determined that for control and structural reasons the pitch-only option was to be developed.

2.2.3 Nozzle Configuration

The nozzle will be of rectangular design for good stealth qualities. The exhaust stream must be shielded from the sides by the tail plane and the nozzle should shield the heat signature from below similar to the YF23. Unlike the YF23 the nozzle should provide pitch-only thrust vectoring.

3 On-Design Sizing

3.1 Air Intake

The process of optimising the intake for design conditions is described in figure 6. It is an iterative process that required feedback from the engine cycle analysis team. The inlet aspect ratio required feedback from aerodynamic (drag) and structures (wings).

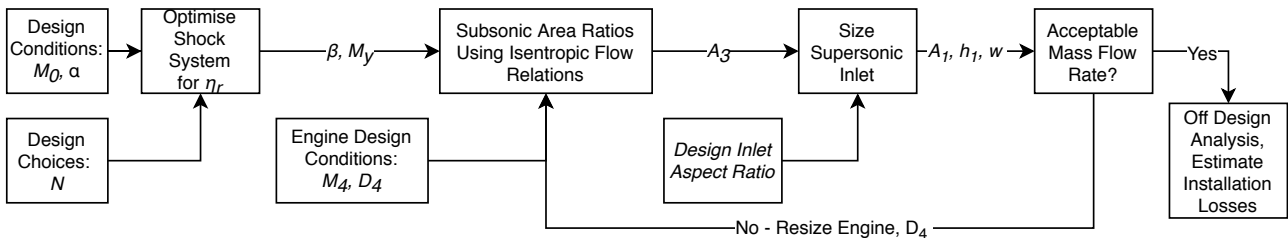
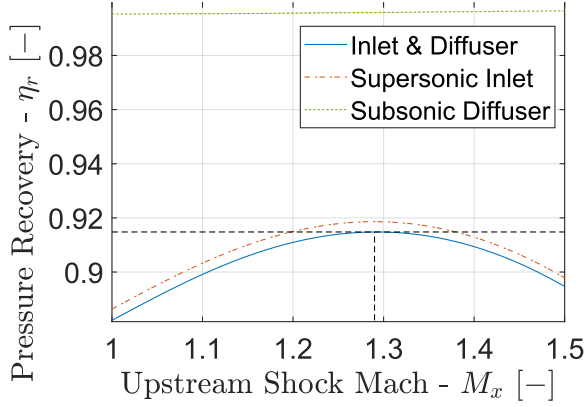


Figure 6: Air intake design process for on-design conditions

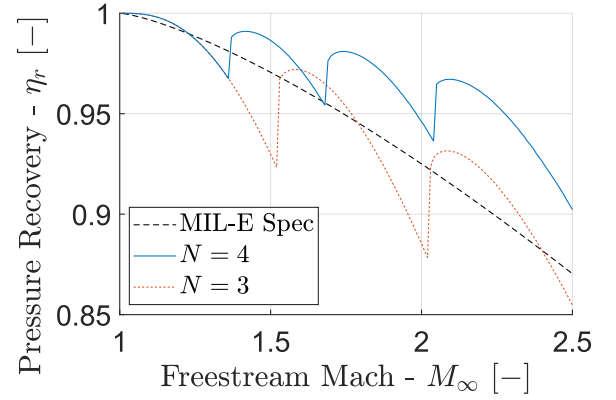
3.1.1 Supersonic Shock System

Supercruise conditions required a mixed compression supersonic inlet (appendix B). A system of three oblique and one normal shock was selected [12] to ensure MIL-E 5007B was met [13]. Figure

7b validates that the four shock system gives acceptable performance across the flight envelope and confirms that a three shock system is insufficient at higher Mach numbers. The poor performance near Mach 1.35 could be improved by using more shocks, however the off-design mass flow ratio discussed in section 4 would be poorer.



(a) Optimal Mach number upstream of terminal normal shock



(b) Pressure recovery for fixed geometry inlet at design point

Figure 7: Optimisation of supersonic shock geometry for total pressure recovery

A multi-variable optimisation program, written using MATLAB attempted to find combinations of ramp angles, which give good pressure recovery at supercruise and combat. It was found that to gain significant improvements at combat required unacceptably large losses in performance at supercruise and a more efficient process for optimising at supercruise was found.

Optimal pressure recovery was obtained by ensuring a constant oblique shock strength and finding M_x such that η_r is maximised (figure 7a). To maintain constant oblique shock strength, equation (1) must be satisfied. The method outlined in [14] is used to identify optimal inlet geometry, the results are presented in table 1.

$$M_n \sin(\theta_{n+1}) = M_{n+1} \sin(\theta_{n+2}) \quad (1)$$

The effect of the Mach cone on M_0 was considered however since the flow undergoes Prandtl-Meyer expansion upstream of the intake, M_0 is found to be very similar to M_∞ .

Table 1: Supersonic inlet geometry and on-design performance

Station		0	1	2	3	4
Area, A	$[m^2]$	0.707	0.707	0.627	0.395	0.395
Height, h	$[m]$	0.523	0.523	0.482	0.293	0.293
Length, L	$[m]$	0	0.417	0.882	1.232	1.732
Ramp Angle, β	$[\circ]$	0	5.58	15.11	5.07	0
Mach, M	$[-]$	2.25	1.92	1.58	1.22	0.83
Pressure Recovery, η_r	$[-]$	1	0.988	0.988	0.988	0.991

In table 1, h is measured from the lower cowl tip and L is measured from the upstream most point of the inlet.

3.1.2 Subsonic Diffuser

Given the engine fan diameter, D_f , desired fan face Mach number, M_6 and the desired inlet throat Mach, M_4 the base geometry of the subsonic diffuser is calculated using isentropic flow relations:

$$\frac{A_4}{A_6} = \frac{A_4 A^*}{A^* A_6} = \frac{1.027}{1.474} = 0.697 \quad (2)$$

From equation (2), inlet throat area is found and exact dimensions of the supersonic inlet are calculated.

The subsonic diffuser must fit around the airframe and other systems in the fuselage and therefore design is restricted. Little research covering optimal serpentine duct design is available. The approach taken was to design a duct that fits the aircraft whilst working with the systems and structures teams and then match the design to a wind tunnel tested duct with good performance.

Table 2: Comparison of subsonic duct geometry with wind tunnel tested serpentine duct

	M_4	$\frac{A_i}{A_4}$	$\frac{A_5}{A_i}$	$\frac{L_{4i}}{D}$	$\frac{L_{i5}}{D}$	$\frac{\Omega_{4i}}{D}$	$\frac{\Omega_{i5}}{D}$
NGF Duct	0.83	1.11	1.33	2.08	1.50	0.75	0.05
Sample Duct [15]	0.6-0.85	1.07	1.30	1.92	1.34	0.86	0.26
Difference [%]	-	3.7	2.3	8.3	11.9	12.8	94.2

Table 2 shows the NGF duct has similar geometry to the sample duct. The designed duct had a higher expansion ratio, but is longer to compensate for this. It had lower flow offset despite still obscuring the fan blades from every angle, which may lead to this duct giving lower flow distortion than the sample duct. Data in [15] suggests this duct gives good pressure recovery, $\eta_{r_{sub}} \approx 0.96$, however flow distortion is an issue. Vortex generators are implemented to bring flow distortion to an acceptable level.

3.2 Exhaust Nozzle

Control of the nozzle throat and exit area is done by a single actuator, reducing weight and complexity however coupling the two areas. Equation (3) expresses the quadratic relation between exit area and throat area where A_8 and A_9 are defined in figure 8a and A_{8d} is the throat area for optimal subcruise performance.

$$\frac{A_9}{A_8} = a + b \left(\frac{A_8}{A_{8d}} \right) + c \left(\frac{A_8}{A_{8d}} \right)^2 \quad (3)$$

$$a = -5.9 \quad b = 6.0 \quad c = 1.2 \quad (4)$$

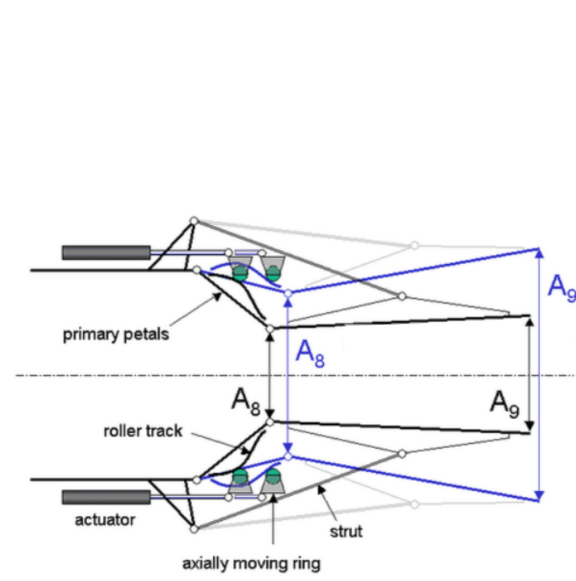
The quadratic can be fitted exactly to three design points, cruise and supercruise cover the largest portion of the mission profile and must have the ideal nozzle settings leaving a third point to be selected freely. Figure 8b shows that optimising for any of the other CMP's gives poor performance at the other two CMP's and requires undesirably large ranges of movement. A polynomial that gives acceptable performance at the remaining CMP's is selected that also limits the required range of motion of the nozzle. Equation 4 lists the coefficients for the selected good trade-off configuration.

4 Off-Design Performance Analysis

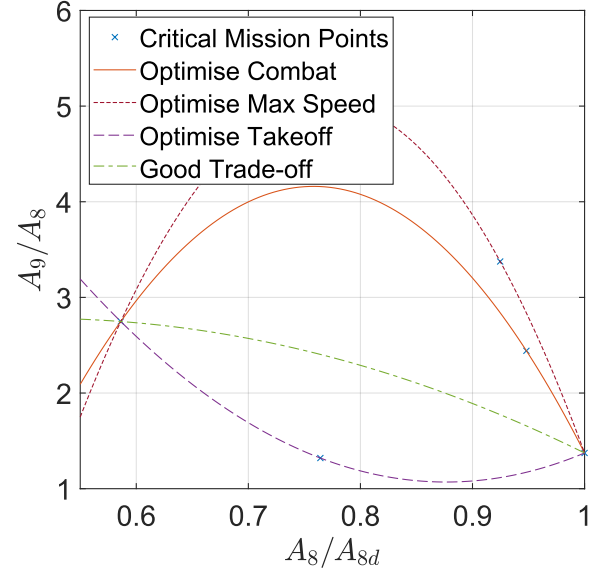
4.1 Air Intake

4.1.1 Area and Mass Flow Ratios

During subsonic flight, air entering the engine may initially accelerate. It is critical that the flow does not reach sonic conditions, [2] as this would lead to supersonic flow downstream of the throat, terminating with a normal shock, giving significant pressure losses. Figure 9a confirms that at the engines current size the flow will not choke for all subsonic flight conditions. Care must be made if the intake is to be made smaller, as figure 9a suggests that a small decrease in intake area could give choked flow. Figure 9b displays the MFR through the intake across the flight envelope. The mass flow ratio is defined as:

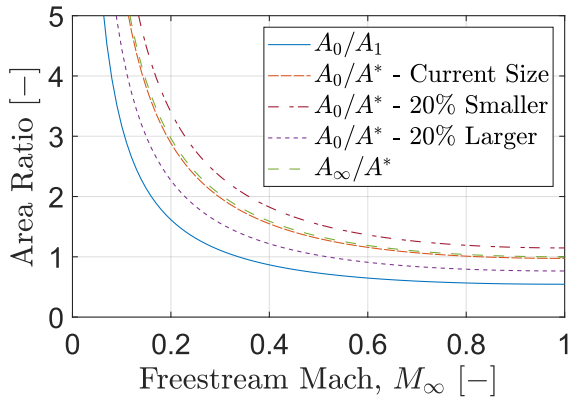


(a) Single actuation, variable geometry nozzle schematic [1]

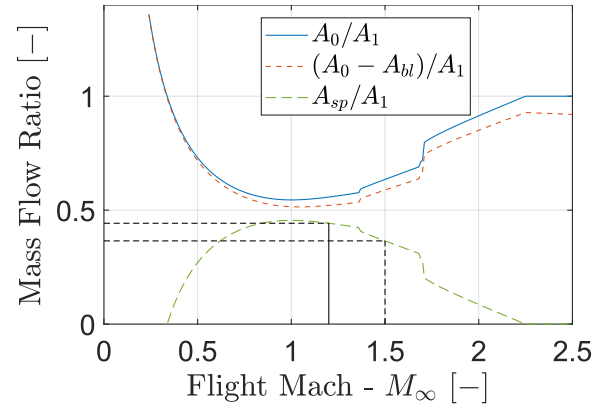


(b) Available variation of area for single actuator variable geometry nozzle

Figure 8: Nozzle configuration



(a) Area ratios at subsonic Mach numbers, fixed altitude



(b) Mass flow ratios at varying Mach numbers, fixed altitude. Showing spillage flow for combat turn and acceleration

Figure 9: Optimisation of supersonic shock geometry for total pressure recovery

$$MFP = \frac{\dot{m}}{\dot{m}_{des}} \quad (5)$$

Where $\dot{m}_{des} = 87 \text{ kg s}^{-1}$ is the design mass flow rate for the engine, provided by the ECA team. The curves show the MFR for the streamtube entering the inlet, the streamtube entering the inlet corrected for boundary layer bleed and the spilled flow streamtube.

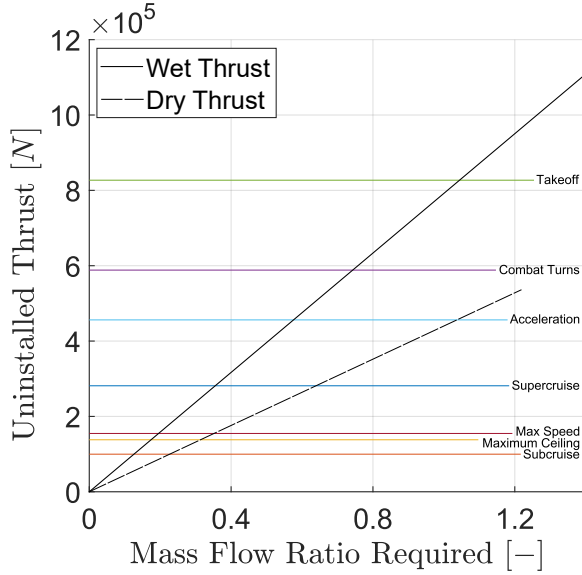
4.1.2 Performance at Critical Mission Points

At the critical mission points, the requirement for mass flow will be highest. Using the constraint diagrams and engine thrust data for wet and dry operation the mass flow ratio required for each CMP is found using figure 10a and listed in table 3. For the first design iteration, it is assumed that:

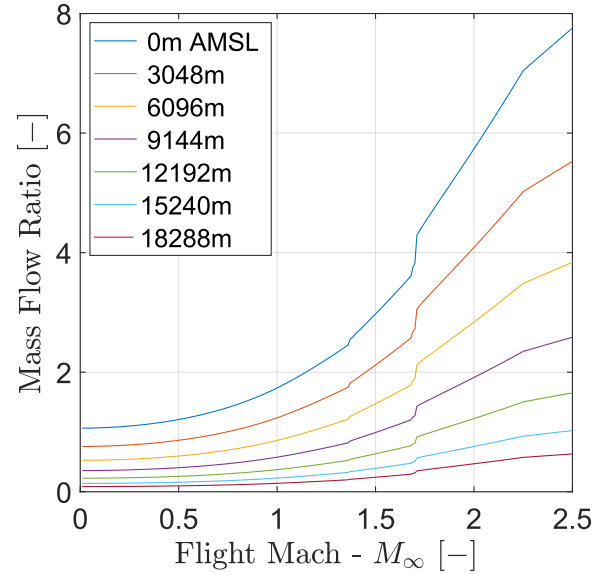
$$\phi_{inlet} + \phi_{nozzle} = 0.09 \quad (6)$$

$$T_{req} = \frac{F_{req}}{1 - (\phi_{inlet} + \phi_{nozzle})} \quad (7)$$

As suggested by [12], where ϕ is estimated in more detail in section 5. The MFR available at different flight velocities and altitudes shown in figure 10b is found using the continuity equation and knowing the area of the streamtube entering the engine from figure 9b.



(a) Engine mass flow required for given thrust setting showing thrust required for critical mission points



(b) Mass flow available to engine at Mach number and altitude (mass flow reduces with altitude)

Figure 10: Optimisation of supersonic shock geometry for total pressure recovery

Table 3 shows a significant MFR deficit for combat conditions. It is expected that the intake will supply insufficient air for takeoff due to low flight speeds. As the intake area is highly constrained and since figure 9b shows significant mass flow spillage at the Mach numbers corresponding to combat turn and acceleration, the option to have a movable lower intake cowl is explored. This potentially removes spilled mass flow at these conditions. With this in mind, a 5% reduction in intake area was considered allowing for a slimmer wing as desired by the aerodynamics (drag) team.

Table 3: Mass flow ratio at critical mission points and intake resizing process.

* - indicates variable geometry intake,

** - indicates resized intake.

Mass Flow Ratio	Supercruise	Subcruise	Max Ceiling	Combat Turn	Takeoff	Combat Acceleration	Max Speed
Required	0.66	0.28	0.33	0.77	1.23	0.59	0.20
Available	0.93	0.30	0.58	0.64	1.09	0.45	0.63
Deficit	-0.27	-0.02	-0.25	0.13	0.14	0.14	-0.43
Deficit *	-0.27	-0.02	-0.25	-0.23	0.14	-0.30	-0.43
Available **	0.88	0.29	0.55	0.60	1.03	0.43	0.60
Deficit **	-0.22	0.00	-0.22	-0.18	0.20	-0.25	-0.40

4.1.3 Bypass Air Flow

In table 3, negative MFR deficits signify the engine receiving excess air which must be vented through bypass doors to prevent supercritical intake operation. During supercritical operation, the terminal

normal shock moves downstream of the throat, giving a stronger shock with a larger total pressure drop which must be avoided.

4.1.4 Auxiliary Air Flow

Auxiliary air hatches will be opened during takeoff to satisfy the engines mass flow demand [2]. For the resized intake, the auxiliary airflow must supply 17.4kg s^{-1} of additional air to the engine.

4.2 Exhaust Nozzle

Table 4 gives the nozzle geometry and flow characteristics at critical mission points. Ideal expansion occurs during sub and supercruise giving best performance for the majority of the flight envelope. At takeoff the exhaust is highly over-expanded whilst during combat it is expected that the exhaust will be under-expanded, both scenarios lead to a loss in performance.

Table 4: Ideal and actual nozzle configuration for critical mission points

CMP	Combat Turn	Max Speed	Supercruise	Takeoff	Subcruise	Min	Max
Ideal Exit Area, A_{9_i} [m^2]	1.13	1.52	0.79	0.49	0.67	0.49	1.52
Actual Exit Area, A_{9_i} [m^2]	0.77	0.80	0.79	0.90	0.67	0.67	0.90
Difference [—]	0.68	0.53	1.00	1.82	1.00	0.53	1.82
Actual Exit Height, h_9 [m]	0.51	0.53	0.52	0.60	0.45	0.45	0.60
Throat Area, A_8 [m^2]	0.46	0.45	0.29	0.37	0.49	0.29	0.49
Throat Height, h_8 [m]	0.31	0.30	0.19	0.25	0.33	0.19	0.33
Convergent Angle, θ [$^\circ$]	6.48	7.34	20.00	13.26	4.59	4.59	20.00
Convergent Length, L_{78} [m]	0.51	0.51	0.51	0.51	0.51	0.51	0.51
Divergent Angle, β [$^\circ$]	8.64	9.94	14.31	15.00	5.17	5.17	15.00
Divergent Length, L_{89} [m]	1.35	1.35	1.35	1.35	1.35	1.35	1.35
Exit Mach, M_9 [—]	2.42	2.76	2.54	1.68	1.74	1.68	2.76

5 Installation Losses

5.1 Air Intake

5.1.1 Subsonic Diffuser

Additive drag, D_{add} accounts for the change in momentum of the streamtube as it transitions from the freestream to the inlet front face and is defined by equation (9) [12].

$$D_{add_{sub}} = P_1 A_1 (1 + \gamma M_1^2) - P_0 A_0 \left(\frac{A_1}{A_0} + \gamma M_0^2 \right) \quad (8)$$

$$\phi_{inlet} = \frac{D_{add}}{F} \quad (9)$$

5.1.2 Supersonic Inlet

During supersonic flight, installation losses are caused by both additive drag and spillage drag [16].

$$D_{add_{sup}} = (P_1 - P_\infty)(A_i - A_0) + (P_2 - P_\infty)(A_1 - A_i) \quad (10)$$

$$D_{spill} = (P_2 - P_\infty)(A_1 - A_0) - D_{add} \quad (11)$$

$$\phi_{inlet} = \frac{D_{add_{sup}} + D_{spill}}{F_{uninstalled}} \quad (12)$$

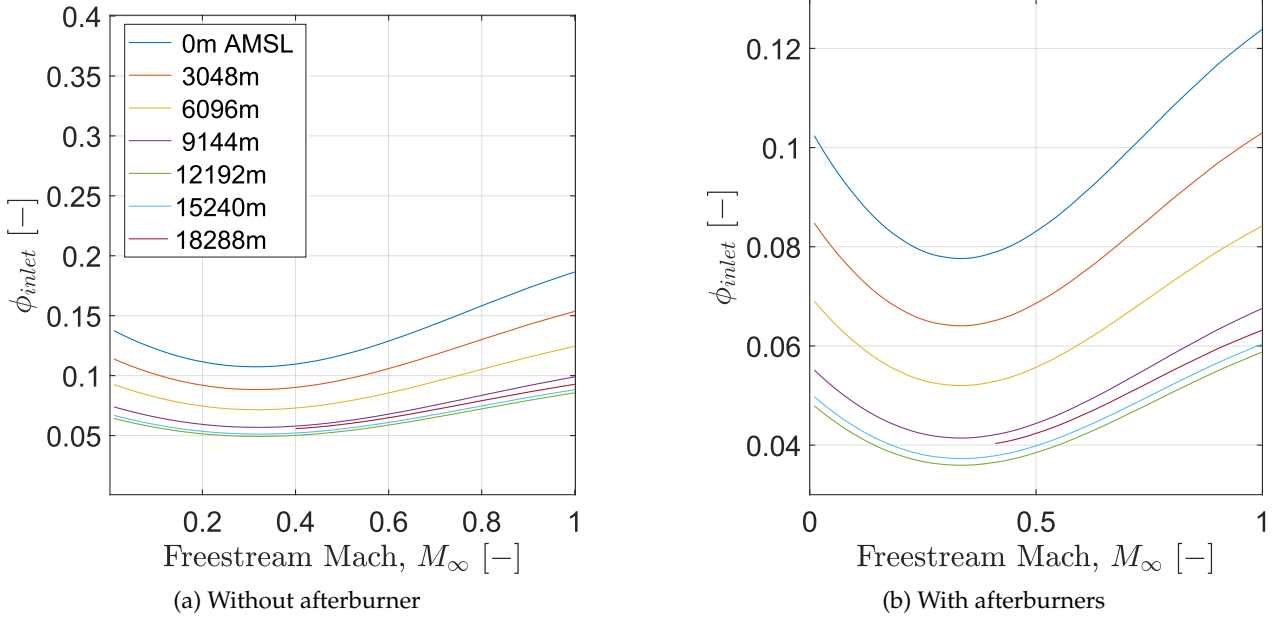


Figure 11: Subsonic and supersonic installation losses

In addition cowl and bypass flow drag contributes to installation losses. [12] accounts for all four sources of supersonic inlet drag with equation (13). The embedded intake gives a reduction in cowl drag as the cowl is part of the wing, reducing the wetted area of the aircraft.

$$\phi_{inlet} = \frac{\left(\frac{A_1}{A_0} - 1\right) \left[M_0 \left(\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} M_0^2 \right)^{\frac{1}{2}} \right]}{\frac{F_g}{m_0 a_0}} \quad (13)$$

It was found that these relations gave unexpectedly large values of ϕ_{inlet} . According to [17], the supersonic mixed compression inlet requires more sophisticated techniques to analyse the installation losses. Instead, the GasTurb [1] software package was used to predict installed engine performance.

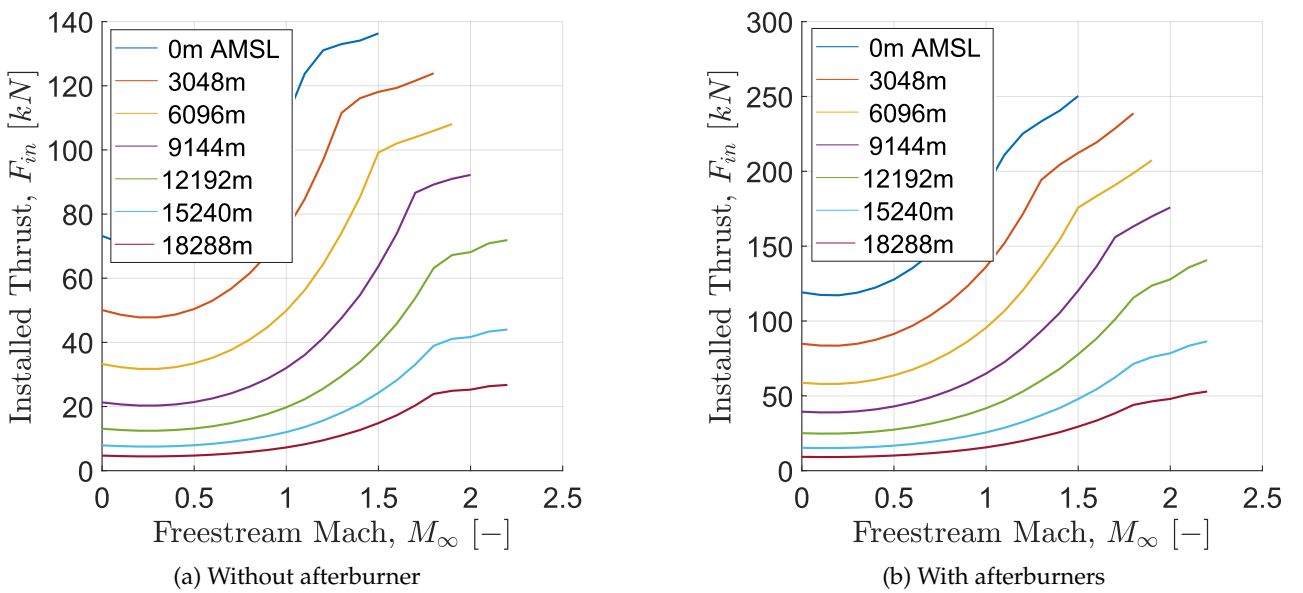


Figure 12: Installed thrust data provided by engine cycle analysis team [18]

Figure 12 shows the installed thrust with afterburner off and on. From this data it was found that the engine is capable of producing just enough thrust to meet the requirements set in the mission profile,

implying that the estimations of ϕ_{inlet} and ϕ_{nozzle} are accurate and the intake is well sized.

5.2 Exhaust Nozzle

5.2.1 Discharge and Velocity Coefficient

The discharge coefficient is a measure of the actual throat mass flow compared to the ideal mass flow without losses. It is defined by equation (15) [2] and is approximately 0.96 for the given nozzle design based on the convergence angle. The velocity coefficient is a measure of how much exhaust flow is slowed by nozzle drag and is estimated by figure 10.61 in [2]. The results are shown in table 5

$$C_D = \frac{\dot{m}_8}{\dot{m}_{8i}} = \frac{P_{t8}}{P_{t7}} \approx 0.96 \quad (14)$$

$$C_V = \frac{V_9}{V_{9i}} \quad (15)$$

5.2.2 Gross Thrust Coefficient

The gross thrust coefficient is the ratio of actual thrust to thrust produced by the nozzle under ideal conditions. It is given by equation (16) [2] and is related to ϕ_{nozzle} . Data for nozzle entry conditions were obtained for the five CMP's considered and the gross thrust coefficient was found to be acceptable for each using the relations above as shown in table 5.

$$C_{fg} = \frac{F_a}{F_i} \quad \phi_{nozzle} = 1 - C_{fg} \quad (16)$$

$$F_a = \frac{\dot{m}_8 V_9}{g} + (P_9 - P_0) A_9 \quad F_i = \frac{\dot{m}_{8i} V_s}{g} \quad (17)$$

Table 5: Calculation of nozzle gross thrust coefficient for critical mission points.

CMP	Combat Turn	Max Speed	Supercruise	Takeoff	Subcruise
C_D	0.960	0.960	0.960	0.960	0.960
C_V	0.997	0.997	0.998	0.998	0.996
C_{fg}	0.990	0.990	0.940	0.950	0.990

These values give $\phi_{nozzle} \approx 1 - 5\%$ which is inline with the estimations made in section 4.

6 Summary of Design

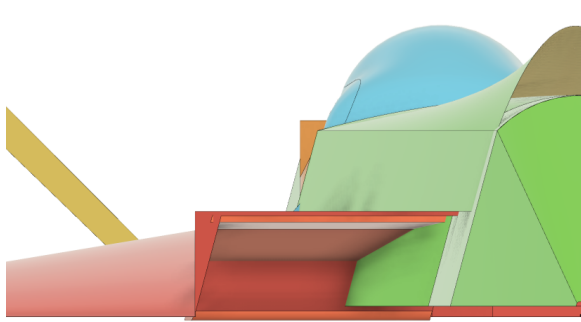
Appendix A provides technical drawings for the air intake and exhaust nozzle components.

6.1 Supersonic Inlet

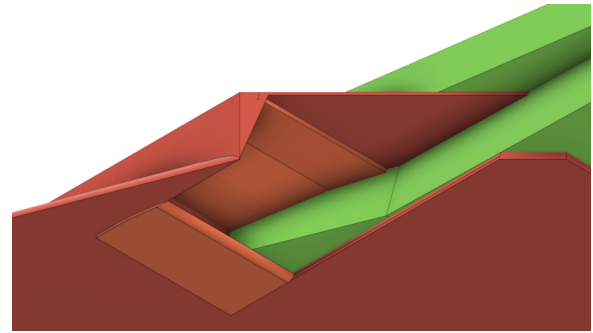
Figure 13 shows the inlet embedded into the wing. These projections do not show where the fuselage must be removed to accommodate the inlet. The figures show the key design features including the swept leading edge and skewed intake face to reduce radar cross section. Also visible is the flat underside of the aircraft that mounting the intakes in the wings provides.

6.2 Subsonic Diffuser

Figure 14 shows the serpentine shape successfully obscures line of sight to the engine fan face, significantly reducing RCS. The duct was generated in CAD by defining the cross section at five evenly spaced points along the duct and performing a loft, ensuring the duct is tangential to the flow at the start and end for 500mm to reduce flow distortion.



(a) Front view

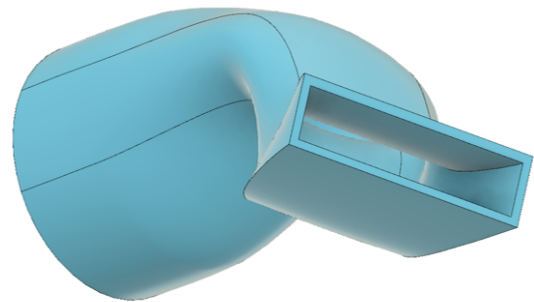


(b) Lower isometric view

Figure 13: Computer aided design projections of supersonic inlet embedded within wing.



(a) Front view

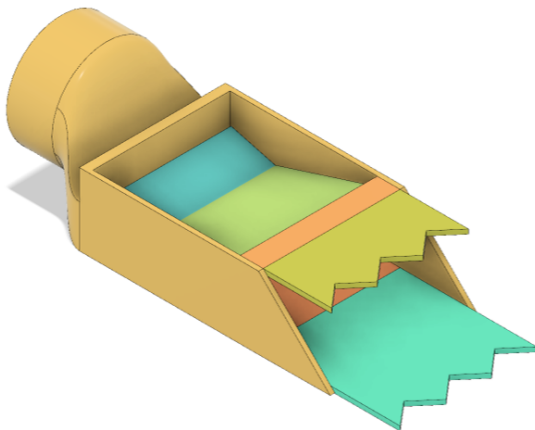


(b) Lower isometric view - showing how the serpentine duct obscures line of sight from every angle

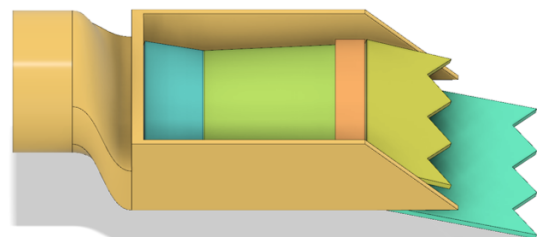
Figure 14: Computer aided design projections of subsonic serpentine duct.

6.3 Exhaust Nozzle

Figure 15 shows the design for the aircraft exhaust nozzle. Note the use of jagged rear edges to deflect radar - these angles match the rear wing sweep to reduce the number of unique angles. The lower part of the nozzle extends further to shield the exhausts heat signature from below. Figure 15b shows the nozzle configured to provide pitch-down thrust vectoring. The nozzle will be covered with a more aerodynamic and stealth cowl. A serpentine duct was also implemented between the turbine and nozzle, allowing the thrust line to be set as desired by the control team and obscuring the turbines from radar, giving better stealth performance.



(a) Isometric view



(b) Side view showing a pitch-down vectoring configuration

Figure 15: Computer aided design projections of exhaust nozzle.

7 Limitations

7.1 Inlet and Diffuser

More research into the effects of a variable geometry intake on spillage would be covered. Mechanisms for actuating the variable geometry, as well as controlling bypass, bleed and auxiliary air flows should be considered, as this would effect the power and hydraulic requirements provided by the systems team. CFD analysis and a review of the impact of vortex generators should validate the proposed design.

7.2 Exhaust Nozzle

The ablative or heat resistant coating applied to the nozzle needs to be considered. The actuation mechanism for throat sizing and thrust vectoring should be considered and passed on to the systems team. CFD analysis should validate the design. It was noted at a late stage in the design process that the mass flow through the nozzle was significantly higher than expected, suggesting that the turbine exit area provided by the ECA team is too large. This should be investigated and the nozzle resized as necessary.

7.3 Installation Losses

The prediction of installation losses is one of the weaker points of this design process. More research is needed into the drag associated with mixed compression inlets.

8 Conclusion

An air intake and exhaust system has been designed for a next generation fighter aircraft. The intake is optimised for supercruise and the exhaust for both super and subcruise. The intake has been sized to match the engine air flow requirements and gives good total pressure recovery across the flight envelope.

A number of stealth features have been implemented into both systems to ensure they are suitable for the next generation fighter aircraft. The boundary layer bleed system in the inlet builds on the advanced technology of the YF-23, giving the aircraft better stealth performance than fifth generation fighters. Embedding the intakes within the wing makes for a more aerodynamic aircraft. The nozzle is able to combine the strengths of both the F-22 and YF-23 by having both thrust vectoring and shielded exhausts whilst also featuring a serpentine duct which is not present on either aircraft.

Performance analysis has suggested both components are suitable for the role although the exhaust nozzle may need significant resizing. These components should be improved upon in the next design iteration. Installation losses will be the area requiring the most refinement whilst the air intake should not need much further refinement beyond considering a variable geometry cowl. More accurate testing techniques such as CFD and detailed installation loss analysis will be performed.

References

- [1] "GasTurb Software", [Online]. Available: <http://www.gasturb.de/>.
- [2] J. D. Mattingly, Elements of Gas Turbine Propulsion. New York; London: McGraw-Hill, 1996, ISBN: 0079121969.
- [3] E. L. Goldsmith and J. Seddon, Practical Intake Aerodynamic Design. Oxford: Blackwell Scientific, 1993, ISBN: 0632031034.
- [4] A. G. Rao and S. P. Mahulikar, "Integrated review of stealth technology and its role in airpower PhD Student , 2 Assistant Professor Department of Aerospace Engineering", no. December 2002, 2018.
- [5] The YF-23's Air Inlet Design Was Its Most Exotic Feature You Never Heard Of. [Online]. Available: <https://www.thedrive.com/the-war-zone/20881/the-yf-23s-air-inlet-design-was-its-most-exotic-feature-you-never-heard-of>.
- [6] "F-22 Inlet", [Online]. Available: <https://cdnph.upi.com/svc/sv/i/9201486049885/2017/1/14860507819912/Lockheed-completes-inlet-coating-repair-on-F-22.jpg>.
- [7] YF-23 Inlet. [Online]. Available: <https://www.flickr.com/photos/vzlet/35736966422>.
- [8] F22 Nozzle. [Online]. Available: <https://i.ytimg.com/vi/ZH1gufgmopA/maxresdefault.jpg>.
- [9] F35 Nozzle. [Online]. Available: <https://i.ytimg.com/vi/q5hBavWq7pU/maxresdefault.jpg>.
- [10] YF23 Nozzle. [Online]. Available: <https://media.defense.gov/2015/Oct/08/2001298489/-1/-1/0/151007-F-DW547-009.JPG>.
- [11] W. Cheng, Z. Wang, L. Zhou, X. Sun, and J. Shi, "Influences of shield ratio on the infrared signature of serpentine nozzle", Aerospace Science and Technology, vol. 71, no. May, pp. 299–311, 2017, ISSN: 12709638. DOI: 10.1016/j.ast.2017.09.001. [Online]. Available: <https://doi.org/10.1016/j.ast.2017.09.001>.
- [12] J. D. Mattingly, Aircraft Engine Design. Washington, DC: American Institute of Aeronautics and Astronautics, 1987, ISBN: 0930403231.
- [13] "Downloaded from <http://www.everyspec.com>", International Journal of Fatigue, vol. 85498, no. February 2004, pp. 136–146, 2018, ISSN: 01421123. DOI: 10.1016/0142-1123(90)90679-9. [Online]. Available: <https://doi.org/10.1016/j.nucengdes.2018.02.039%7B%5C%7D0Ahttp://linkinghub.elsevier.com/retrieve/pii/S2452321616304309%7B%5C%7D0Aftp://185.72.26.245/Astm/1/Section%2003/ASTM0301/PDF/E1049.pdf>.
- [14] H. Ran and D. Mavris, "2005, Ran, Preliminary design of a 2d supersonic inlet to maximize total pressure recovery.pdf", AIAA Paper, no. September, pp. 1–11, 2005. [Online]. Available: <http://arc.aiaa.org/doi/pdf/10.2514/6.2005-7357>.
- [15] S. Sun and R. W. Guo, "Serpentine inlet performance enhancement using vortex generator based flow control", Chinese Journal of Aeronautics, vol. 19, no. 1, pp. 10–17, 2006, ISSN: 10009361. DOI: 10.1016/S1000-9361(11)60261-6.
- [16] J. C. Esterhuysen, "Aerodynamic Drag of a Two-Dimensional External Compression Inlet at Supersonic Speed", 1997.
- [17] J. Roskam, Airplane design. Part 6, Preliminary calculation of aerodynamic, thrust and power characteristics. Ottawa: Roskam Aviation and Engineering Corporation, 1987, ISBN: 1884885098.
- [18] M. Fok, "Next Generation Fighter, Individual Reprot - GDP 2019", 2019.
- [19] External Compression Inlet. [Online]. Available: <https://rocketgirl2017.files.wordpress.com/2018/05/rampshockschem.png?w=964>.
- [20] Mixed Compression Inlet. [Online]. Available: <https://i.stack.imgur.com/SJh0h.png>.

A Technical Drawings

All dimensions are in mm. A brief description of the technical drawings presented:

A.1 Supersonic Inlet

Presents a three view projection of the supersonic portion of the inlet responsible for bringing flow to subsonic speeds. Angles are optimised for $M_0 = 2.25$.

A.1.1 Side View

A more detailed side view of the supersonic inlet showing lengths and heights of each ramp system.

A.2 Subsonic Diffuser

Shows the serpentine duct used to slow subsonic flow to the ideal engine face Mach number (M_6).

A.3 Full Inlet

Shows one supersonic inlet and subsonic diffuser connected as they would be when integrated with the aircraft.

A.4 Exhaust Nozzle

A.4.1 Three View Drawing

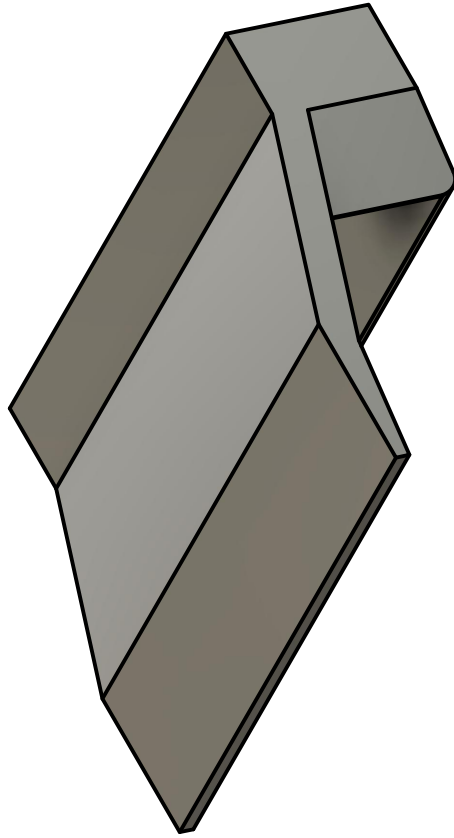
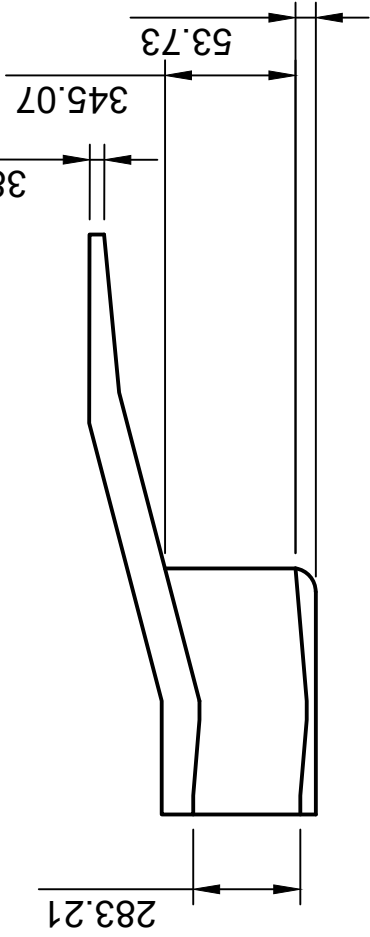
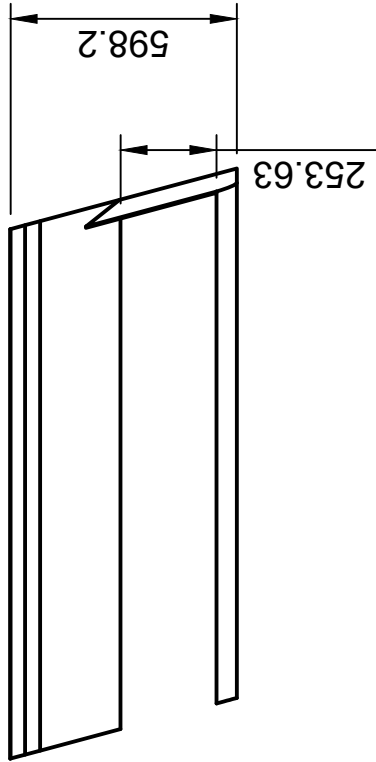
Three view drawing of exhaust nozzle with wide throat setting, suitable for afterburner on case. Following page shows a sectional view of the convergent and divergent ramps.

A.4.2 Sectional View: Narrow Throat Setting

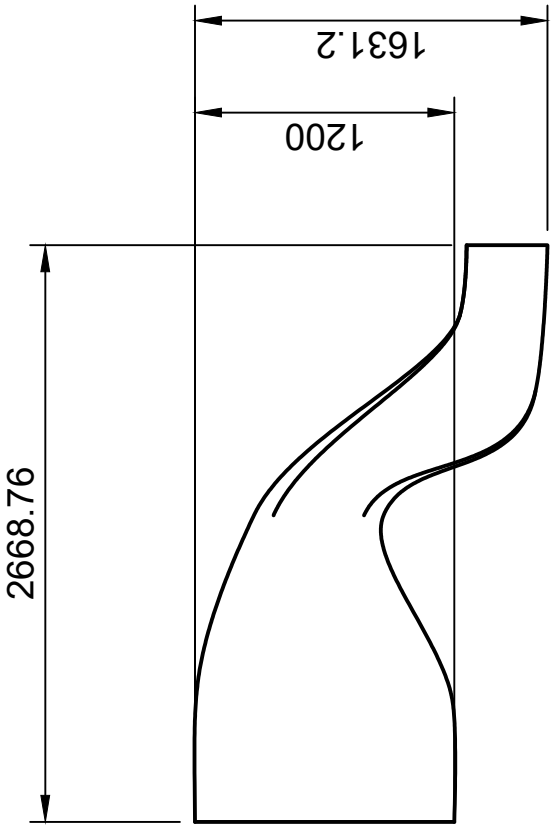
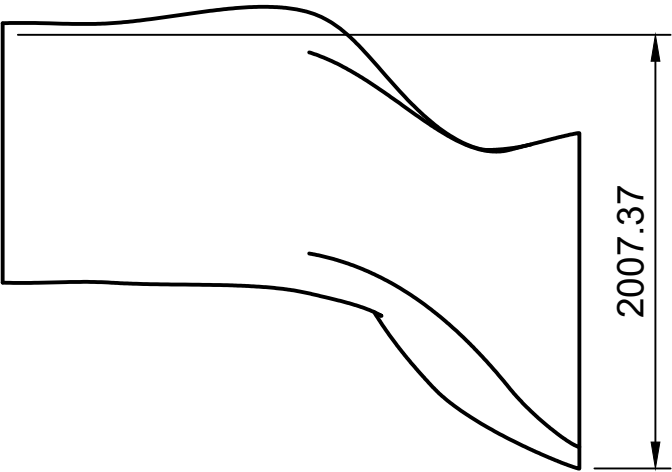
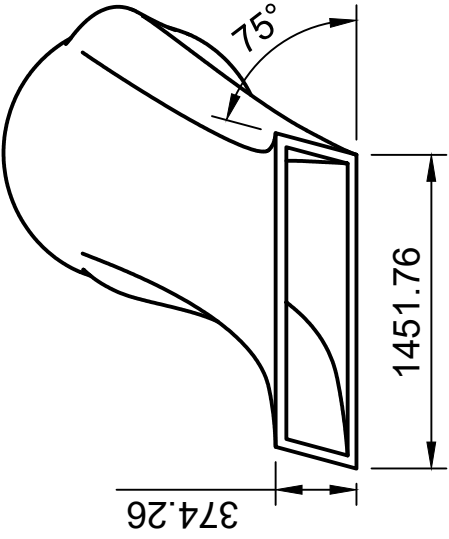
Sectional view showing convergent and divergent ramps with narrow throat setting, suitable for cruise/supercruise.

A.4.3 Sectional View: Thrust Vectoring

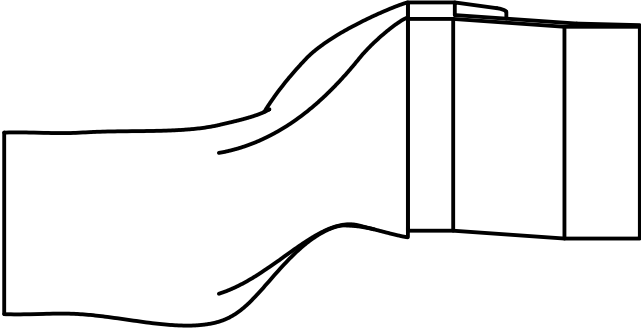
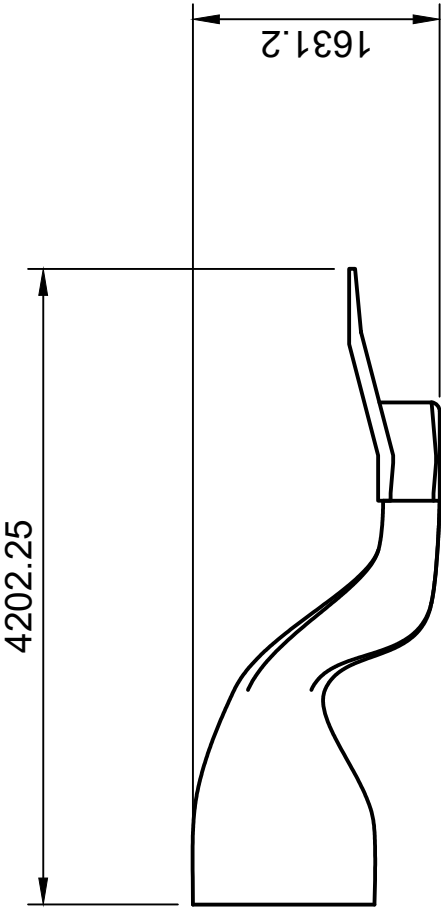
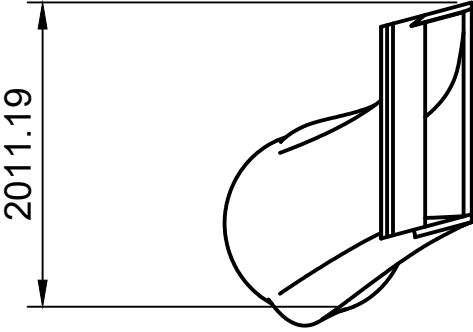
Shows how nozzle is positioned for a pitch down thrust vectoring maneuver.



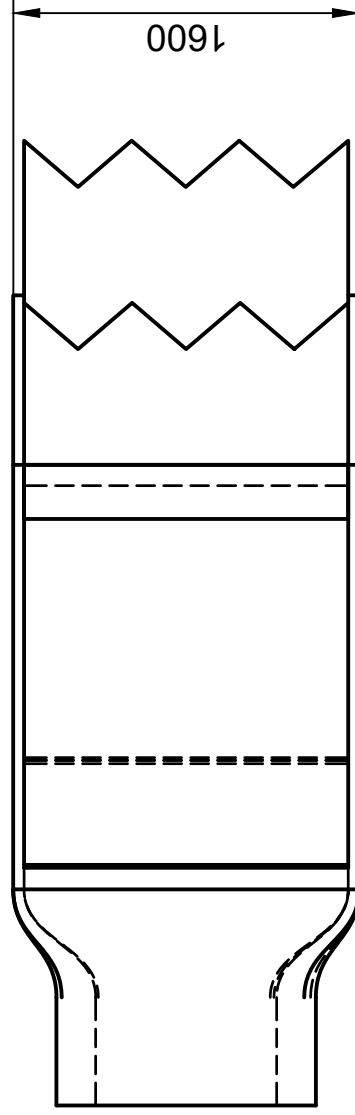
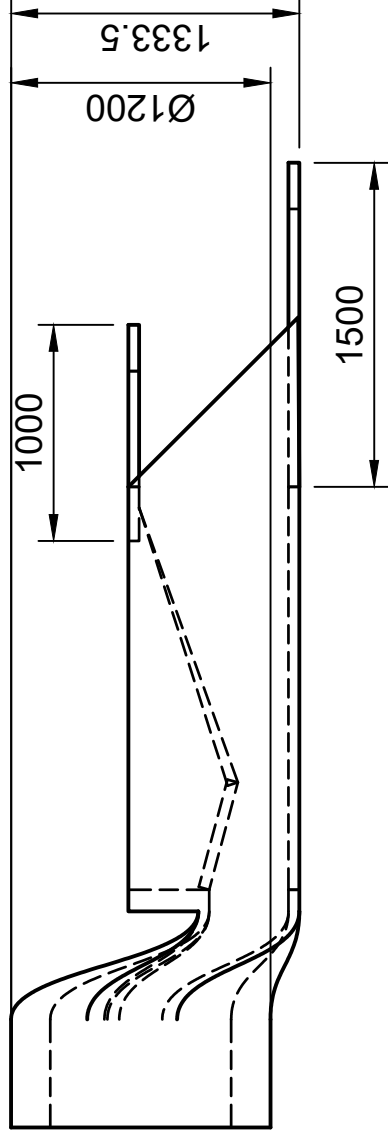
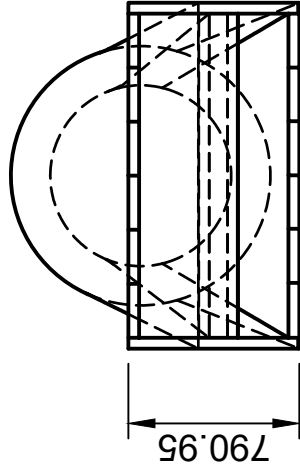
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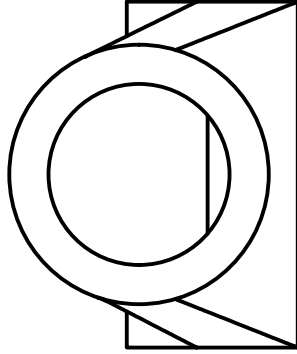


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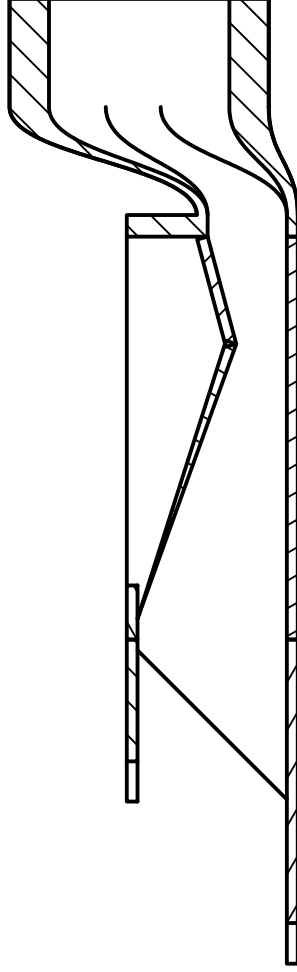


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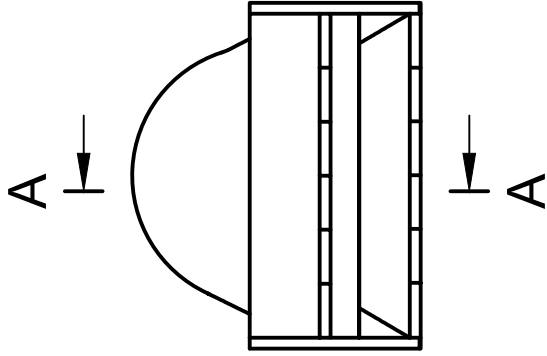
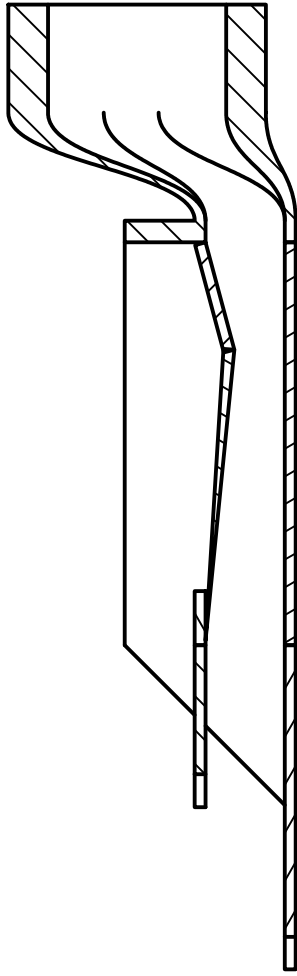
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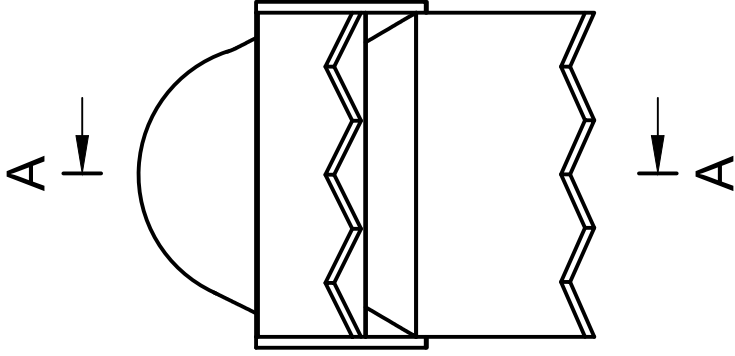
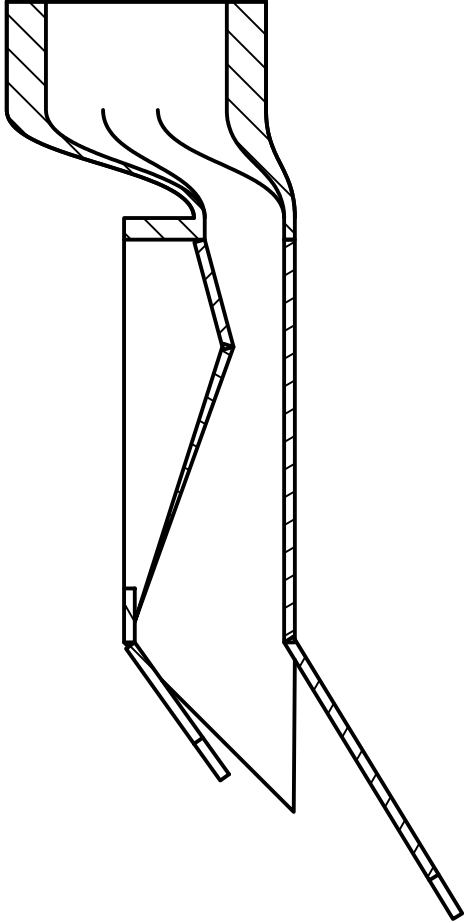
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			Rev.	Date of issue	Sheet 1/1



Dept.	Technical reference	Created by Andrew Bates	17/06/2019	Approved by			
		Document type		Document status			
		Title Exhaust Nozzle Pitch Down	DWG No.				
				Rev.	Date of issue	Sheet 1/1	

B Supersonic Inlet Configurations

The pitot intake in figure 16 is the most basic supersonic intake and works by generating a weak normal shock just inside the cowl. The shock must be swallowed by overspeeding or increasing airflow through the engine. For these reasons and because the normal shock induces heavy pressure losses at high Mach numbers; the pitot inlet is only suitable for flight below Mach 1.6.

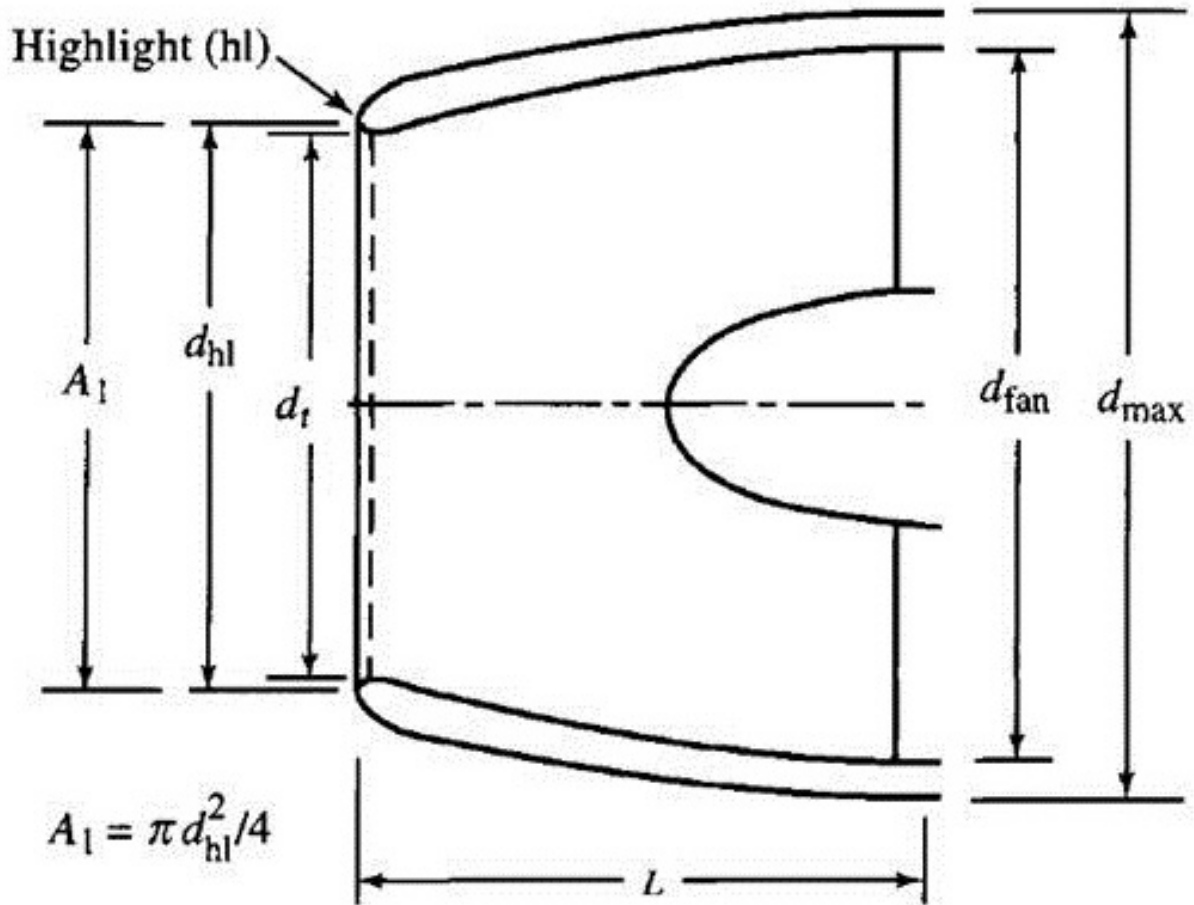


Figure 16: Pitot supersonic intake [2]

The external compression intake (figure 17) uses a series of weak oblique shocks followed by a normal shock to bring flow subsonic. The weaker shocks induce lower pressure losses making the inlet more efficient. Above Mach 2.2 the external inlet geometry becomes large and induces large drag.

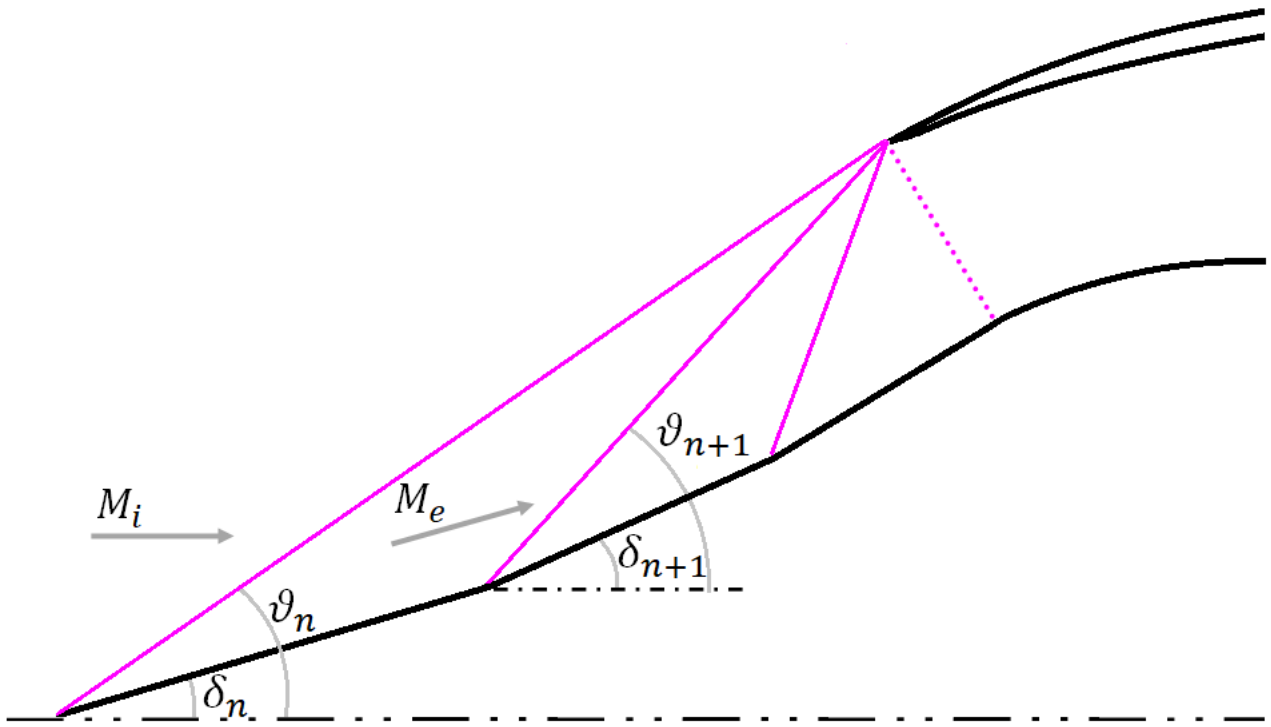


Figure 17: External compression intake [19]

Above Mach 2.2 the mixed compression inlet (figure 18) is most suitable. It is similar to the external compression inlet however some of the oblique shocks occur internally. This allows for a smaller intake reducing drag however these inlets are more complicated to design and operate.

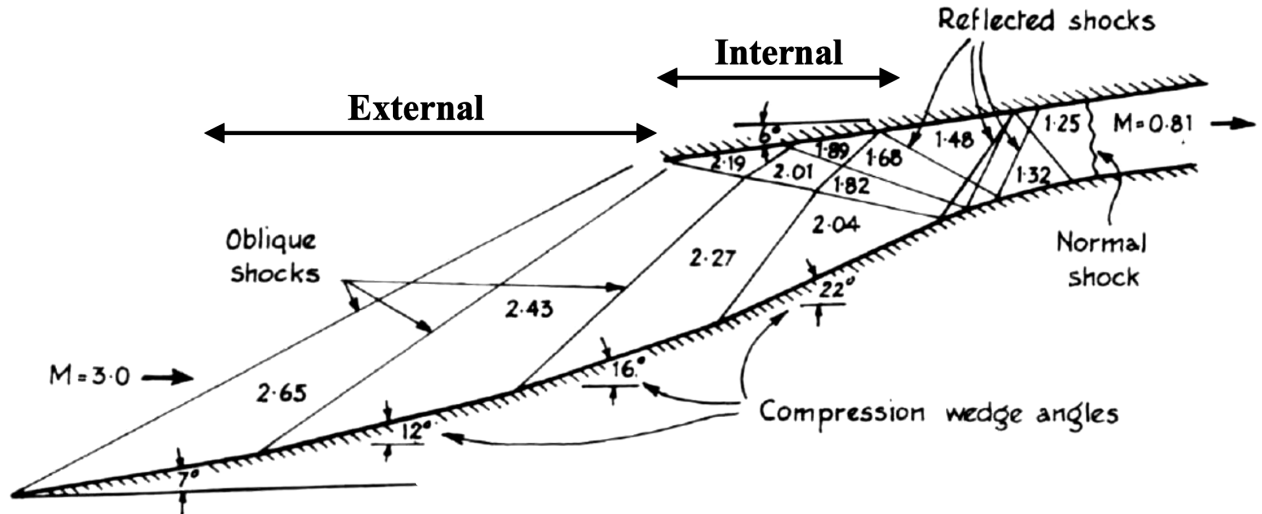


Figure 18: Mixed compression intake [20]